

Keep Your Distance: A Cognitive Approach to Mobile Robot-Cyclist Interaction

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Abstract—This paper proposes a framework for integrating the subjective safety of vulnerable road users into autonomous driving (AD) systems' decision-making. Cognitive-psychological data on cyclists are collected in laboratory experiments and validated in an empirical field test. The identified subjective safety parameters ensure the optimal distance and velocity for AD systems, represented by a mobile robot. Approaching decision-making design from a human-centred perspective ensures safety, comfort, and high acceptance of AD systems.

Index Terms—Autonomous Mobile Robots, Traffic Safety, Human Factors

I. INTRODUCTION

With growing interest in autonomous driving (AD) technologies, discussions around trust, ethics, and safety regarding their integration into daily life are increasing. Trust and acceptance of the technology depend on the perception of safety. In road safety, there is a distinction between objective and subjective safety. Objective safety reflects how unlikely it is for a road user to be involved in an accident [2]. The lower the probability of an accident, the higher the objective safety. It is determined by observable data such as accident statistics, conflict analyses, and infrastructure quality [3]–[5]. Subjective safety, on the other hand, refers to road users' individual perceptions of their safety in road traffic, regardless of the likelihood of an accident [2], which is evaluated with surveys or psycho-physiological measurements [6], [7]. Both objective and subjective safety interact closely.

The presented paper researches subjective safety in the interaction between vulnerable road users (VRUs) and AD systems in urban areas with empirical laboratory and field experiments. Specifically, the focus is on cyclists, who are part of the non-motorised vulnerable road users, following the taxonomy of [8]. A last-mile delivery robot represents

the AD system. Increase in cycling and autonomous mobile robots (AMRs) application offer an economic and ecological solution to urban logistics and mobility challenges, improving quality of life by reducing road traffic [9] and improving customer-centric services, such as last-mile deliveries. To build a robust AD system, the level of trust in AD systems, the adaptability, and the situation interpretation processes should be increased by linking cyclists' subjective safety criteria with the decision-making system of AD technologies.

A. Objectives

VRUs and AD systems have an increased potential for conflict on urban roads and areas [1], [10], therefore, decision-making in view of interaction with VRUs is crucial for safe driving and high acceptance of AMRs in traffic. As most research in decision-making for AD focuses on vehicle-to-vehicle interaction [11], this research intends to expand the knowledge on the human-centred perspective and proposes the integration of human factors into the decision-making. The general objective is the exploration of the relationship between cyclists' subjective safety and behavioural parameters of the AMRs on cycle paths in urban environments, in the scope of distance and velocity configuration. This work proposes the methodology of a framework for integrating subjective safety criteria in cyclist-robot interaction. The novelty is in its dense focus on the particular VRU group's interaction with the AD system.

II. RELATED WORK

For the development of the framework, it is important to understand the influences on cyclists' subjective safety and the role of AD in this context. Reference [12] states that factors such as existing infrastructure, traffic volume, and the behaviour of other road users influence subjective safety alongside personal experiences [13]. In the domain of pedestrian-AD vehicle interaction, [14] highlight that human

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factors are a crucial yet under-explored aspect of AD research. Their study concluded that AD vehicles invoked higher gap acceptance times as well as stress levels of pedestrians compared to non-AD vehicles, indicating difficult interactions and lower trust. Research specifically on cyclist-AD system interaction is limited. Reference [15] investigated differences in attitudes and trust between two groups of cyclists with different levels of interaction experience with an automated shuttle bus (ASB) in shared areas. It was identified that the incidents, which respondents mentioned, typically involved harsh braking on a crowded walkway, potentially indicating that the cyclists did not expect the buses to brake, an intention recognition issue. Regarding cyclist-AMR interaction, identifying human factors of subjective safety becomes more challenging due to complex social scenarios, as the Operational Design Domain in urban areas is less structured and unpredictable. Since no specific studies were found on cyclist-AMR interaction, the literature review focused on human-robot interaction (HRI) regarding proximity theory to transfer the findings to cyclist-AMR interaction. While attempts were made to apply human social norms to HRI [16], [17] noted that methodological and theoretical differences limit direct comparisons, highlighting a research gap. Reference [18] identified two key HRI scenarios on sidewalks: the (non-predictable) cornering scenario and the (predictable) direct encounter. The study found that pedestrians preferred cautious robot behaviour in unpredictable situations but favoured faster movement when the robot's path was clear and predictable.

Based on the related work, it is hypothesised that *cyclists feel less safe when robots are close and moving fast in non-predictable scenarios. Conversely, slower speeds and greater lateral and longitudinal distances enhance subjective safety in such scenarios.*

III. METHODOLOGY

As discussed in Section I-A, integrating subjective safety factors into the AMRs' decision-making system requires defining distances and velocity configurations which are perceived as safe by cyclists. The following scenario is a first application scenario for the proposed framework. It is used to test and prove the methodology. If proven successful, the experiments described in this section need to be repeated with further scenarios, as described in Section V.

A. Scenario

Autonomous delivery robots that use the cycling infrastructure are confronted with a variety of different types of cyclists. This includes road users who operate at low speeds or lack the confidence to handle a bicycle, such as older people or children. The upcoming studies will investigate the perception of cyclists in interaction with a robot that features varying behaviour. The overall objective is to ascertain which setting, in terms of distance and velocity, offers the optimal level of subjective safety for the cyclist.

1) Scenario Description: On a German cycle path, an AMR and a cyclist move in parallel. As described in Section I, the AMR is a stand-in for AD systems. The AMR is on the left side of the path, and the cyclist is on the right side. Both start from the same position and have the same velocity. The cyclist keeps a constant velocity throughout the scenario. No additional traffic rules apply, and no other vehicles are involved, neither in the same nor the opposite direction. The AMR needs to reach a point on the right edge of the path, 50 m from the starting point. At the target point, it needs to come to a full stop, analogous to stopping for a delivery or parking. The AMR is capable of travelling up to 20 km/h. Speed differences of 1:1.2 (slow), 1:1.8 (medium), and 1:2 (fast) will be tested. As a distance constraint, the Proxemic theory [19] is utilised. Humans have four categories of social relationships reflected in social distances. The following distances will be tested in the scenario:

- personal distance (0.45 m – 1.20 m): physical contact between close friends, as well as everyday interaction among acquaintances;
- social distance A: (1.20 m – 2.00 m): business-like social interaction;
- social distance B: (2.00 m – 3.60 m): formal-like social interaction.

The intimate and public distances are not considered due to safety concerns and the width of German cycle paths.

B. Empirical Data Collection

The hypothesis reflected in Section II is being examined in the data collection studies. The design, i.e., setting, population, and analysis, is determined by this hypothesis. Within the framework, virtual reality (VR) laboratory studies are planned to be conducted alongside a field test validation. Distance and velocity configurations are compared in a within-subject setting.

Multiple data sources, both quantitative and qualitative, are used to analyse the cyclists' subjective safety. The different measures are synchronized using time stamps. In the framework, it is hypothesized that a low level of subjective safety induces stress and correlates with externally measurable central nervous activation [20]. Eye-based measures for quantifying central nervous activation include blink rate [21], gaze direction, dwell times, and fixations [22] from which visual areas of interest can be derived [23]. Mobile eye-tracking glasses record these measurements with the help of reflecting infrared light. The measures heart-rate variability (HRV) and skin conductance provide similar data on central nervous activation. Low variability in heart rate, both in frequency and time, indicates higher activation [24]. The heart rate can be recorded with a photoplethysmography sensor attached to a finger or an ear lobe. With central nervous activation, sweat glands produce more sweat, measurable as increasing skin conductivity [25] using electrodes attached to two fingers. Skin conductivity has a tonic component (skin conductance level — SCL) and an event-related component (skin conductance response — SCR). Behavioural measures, such as steering and braking behaviour, quantified by reaction

times and accuracy rates [26], [27], reveal cognitive processing demands [28]. To collect subjective measures such as the perceived level of safety [29], a questionnaire is used. Furthermore, cycling expertise, sense of safety, and qualitative impressions of behaviour are surveyed to assess safety perceptions, validate psycho-physiological data, and support cross-validation for detecting subtle changes in nervous system activation. Confounding variables are countered with baseline measurements. Moreover, video recordings are used to filter out measurements caused by physical motion (cf. Section III-C). Consistent lighting conditions are ensured in VR, although no pupillometric measurements are performed.

1) *Study Procedure*: Two types of VR experiments are planned. First, real-world VR, similar to [30], will be used as proof-of-concept for the data collection. In real-world VR, conditions are filmed with a 360°-camera and replayed via a VR headset. The participant cannot move but only look around by turning their head. The conditions' recordings can be repeated until they meet the study's requirements. All participants observe the identical conditions. The fast and cheap implementation makes it ideal for prototyping.

In the second experiment, the bike simulator at Bauhaus-Universität Weimar will be used. The software replicates a real-world location with traffic in VR [31]. The hardware available at Bauhaus-Universität Weimar is a static base simulator with steering and speed sensors, visualizing via a head mounted display (HMD), comparable to [32]. For both approaches, the AMR model will be recreated by the defined scenario described in Section III-A1, with the appropriate behavioural parameters. Based on the findings of the initial study, the study design is refined as required, and the feedback of the participants is taken into account. In comparison to the real-world VR study, the cyclists' behaviour is also examined. The obtained parameters will serve as the acceptance constraints for the decision-making layer.

The empirical phase concludes by validating the results from the laboratory tests in a field study. The participants will interact with an AMR that follows the identified optimal behaviour parameters. The validation study will show whether the parameters are applicable in the real world or whether there may be other, previously unconsidered influences that require further investigation.

As there are three distances and three speeds defined in Section III-A, the studies all will have a 3x3 design with an additional baseline condition. The order of conditions will be randomized for each participant. There will be separate cohorts for the studies in order to prevent any interaction of effects, though all have the same requirements to the participants. When recruiting, a gender balanced ratio is ensured. The minimum age is 18 years to be able to give consent. Participants must have normal vision or corrected-to-normal vision with contact lenses for comfortable use of the HMD and the mobile eye-tracking glasses. In addition, participants must, based on self-assessment, be able to ride a bike safely and must not have any health conditions that could interfere with the physiological measures, nor should they

be taking any medications that could interfere. Throughout experiments, participants will be provided informed consent prior to participation, and all personal data will be anonymised. Safety protocols will be followed to minimise risks, with robots operated under controlled conditions to ensure participant well-being.

C. Data Analysis

The collected data is analysed, following the general steps of empirical research. For the qualitative data, a qualitative content analysis according to [33] is performed. The quantitative data must first be cleaned of artefacts and outliers, using the mean absolute deviation (MAD) approach [34]. Filming participants during data collection allows filtering unwanted, motion-related peaks, as practised by [14]. For better comparability of the data between participants, it is normalised with baseline data. To accept or refute the hypothesis, correlations, tests of significance and post-hoc tests are calculated.

For the decision-making system, we select the set of parameters with the highest perceived level of safety and lowest level of arousal, on a mean average.

D. Decision-Making System

The presented research adopts the architecture described by [35] for an important factor: the architecture incorporates any available AD system functionality that is currently in development or might be developed in the future. The architecture can therefore be implemented for any type of AD system. The given approach ensures that the AD system's performance and automation level increase continuously over time. The architecture presented by [35] consists of five clusters. *Perception* is responsible for perceiving the environment and generating raw sensor data. The *Situation Interpretation* cluster translates the environmental information into an environmental model, which is later fed to the *Decision-making and Planning* cluster. The *Decision-making* cluster generates the optimal path from this data. The *Tactical Planning Layer* of the *Decision-making* cluster reasons actions or manoeuvres to generate motion specifications that follow the global goal and current situation. The *Trajectory Planning* cluster transforms the motion specification into trajectories, which serve as input for the *Vehicle Dynamics Control*. The *Vehicle Dynamics Control* cluster generates control variables such as steering angle and acceleration.

This framework focuses on the Tactical Planning Layer of the Decision-making cluster for cyclist-robot interaction in urban areas. The generic prototype of the Tactical Planning is illustrated in Fig. 1. It operates hierarchically, ensuring a structured approach to decision-making. Behavioural tasks such as crossing, adjustments to subjective safety constraints, and overtaking slow VRUs (dynamic obstacles) will be evaluated and executed using a Reinforcement Learning (RL) paradigm. This allows for adaptability and flexibility when responding to unpredictable and uncertain situations. To adjust to subjective safety expectations during the interaction with VRUs, the Constrained Soft Actor-Critic (C-SAC) [36] algorithm will be used due to a hybrid discrete-continuous action.

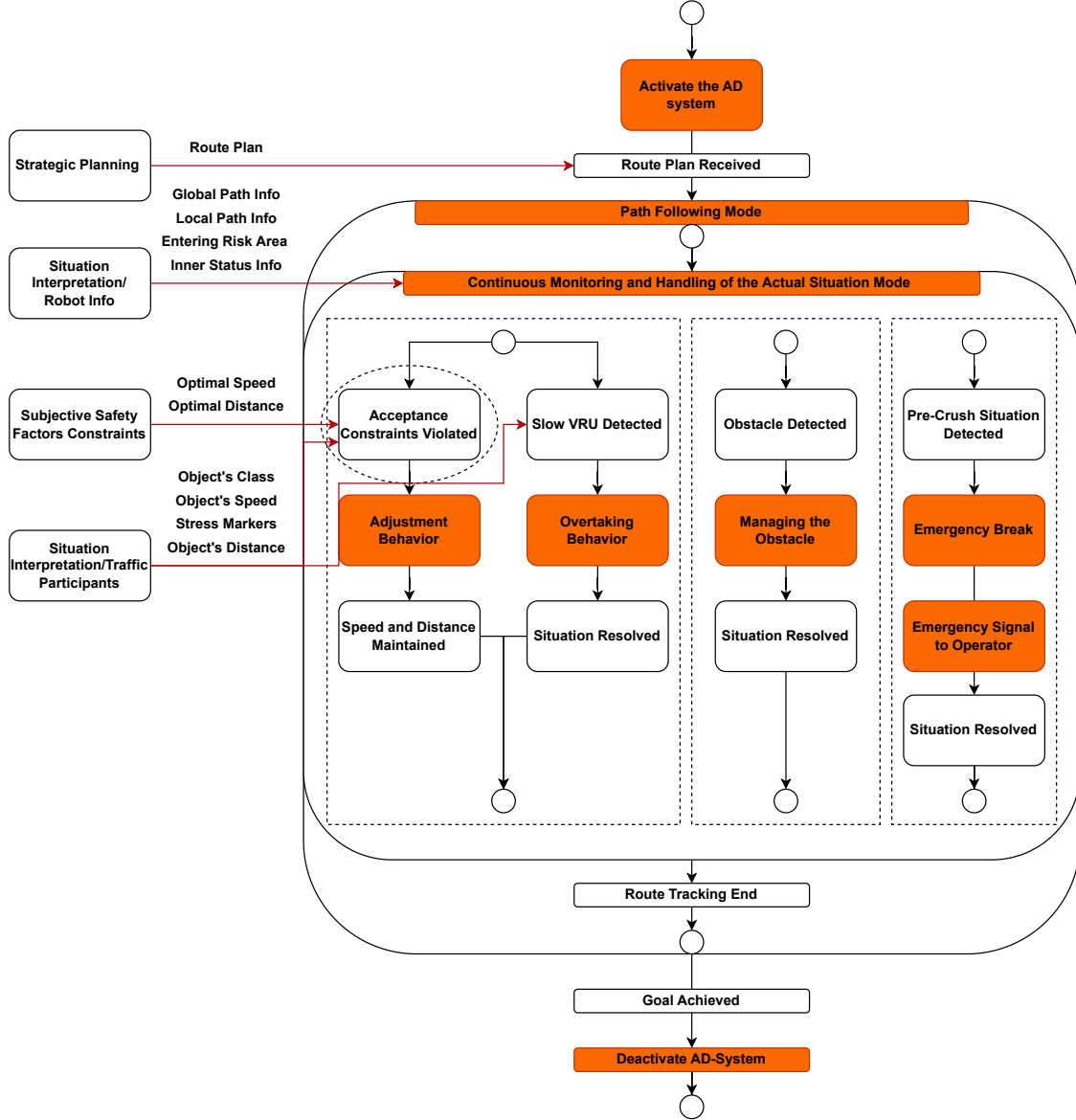


Fig. 1. Tactical planning conceptualisation for the formulated scenario and possible conflict zones and situations.

Discrete actions such as manoeuvre selection (slow down, speed up, overtake, etc.) and continuous actions such as speed control, including space with safety constraints, obtained in a subjective safety investigation, are orchestrated. Therefore, the observed subjective safety factors, like preferred speed and distance, can be integrated as constraints.

$$c_{\text{dist}}(s) = \max(0, d_{\min} - d_{\text{human}}(s)) \quad (1)$$

where:

- $c_{\text{dist}}(s)$ is the constraint cost at state s , penalizing proximity to a human.
- $d_{\text{human}}(s)$ represents the Euclidean distance from the robot to the closest human at state s .

- $d_{\min}$ is the subjective safety distance obtained from an empirical study described in III-B that must be maintained to ensure comfortable interaction. For the identification of the acceptable distance, subjective reports will be prioritised over physiological markers. Later $d_{\min}$ should be normalised.

As a soft constraint for the speed $v_{\max}$, the following smooth logistic cost function will be implemented:

$$c_{\text{speed}}(s) = \frac{1}{1 + \exp(-k_2 \cdot (v_{\text{robot}}(s) - v_{\max}))} \quad (2)$$

where:

- $c_{\text{speed}}(s)$ is the constraint cost at state s related to the robot's speed.
- $v_{\text{robot}}(s)$ is the scalar speed of the robot at state s .

- $v_{\max}$ is the subjective safety speed limit obtained from an empirical study.
- k_2 is a positive scaling factor that controls the steepness of the transition around $v_{\max}$.

This logistic function smoothly increases the penalty as the robot's speed exceeds $v_{\max}$, making it suitable for gradient-based policy optimisation. Therefore, the subjective safety parameters will be integrated via constraints into the objective of the C-SAC [37] approach as follows:

$$\mathcal{L}(\pi, \lambda) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot | s_t))) \right] - \sum_i \lambda_i \left(\mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t c_i(s_t, a_t) \right] - d_i \right) \quad (3)$$

where:

- π is the stochastic policy.
- $\mathcal{H}$ is the entropy of the policy to encourage exploration.
- λ_i are Lagrange multipliers for constraints.
- $c_i \in \{c_{\text{dist}}, c_{\text{speed}}\}$ are constraint cost functions.
- d_i are the upper bounds for expected cumulative violations.
- γ is the discount factor.

This formulation ensures the policy maximises reward while respecting safety constraints by penalising cumulative violations exceeding their respective thresholds d_i . The constraints are formulated to encourage flexibility and smoother learning. Meanwhile, critical safety functions, including Emergency Stops and Obstacle Management, when the obstacle is static, will be tackled using a Finite State Machine (FSM) to ensure strict safety control, playing a safety shield role. This is reflected in Fig. 1, RL learns to satisfy constraints globally and optimise performance, and FSM guarantees constraint satisfaction locally and handles edge cases. Together, they provide both learning-based adaptability and rule-based safety guarantees. All subsystems related to manoeuvring are integrated within the adjustment behaviour and slow VRU avoidance cluster, as shown in Fig. 1. The proposed concept of the Tactical Planner is closely aligned with cyclists' subjective safety perception, ensuring behaviour that meets human expectations and comfort. In this paper's preliminary stage, the factors considered are optimal speed and distance. However, it is designed to be extendable to additional criteria and behaviours. This framework incorporates the importance of linking subjective safety factors with an AD system's decision-making system, reflected in Section I.

IV. EXPECTED OUTCOMES

The presented paper is the preliminary step of assessing the methodology and theoretical approach. All proposed measures are currently still ongoing or in a planning stage. The proposed approach is intended to retrieve statistically significant parameters for velocity and optimal distance for AD systems. The implementation aims for an improved decision-making system that increases the (subjective) safety of VRUs.

It is planned to obtain a unique dataset of cyclist-robot interaction, which researchers can use for training stress recognition or further cyclist-robot interaction models. Increasing the subjective safety enhances public trust and acceptance of AD systems, which in turn supports wider societal adoption [38]. Greater implementation of AD technologies can have multiple positive effects, not only reducing traffic accidents but also benefiting the environment and improving quality of life. For example, in the case of last-mile delivery robots discussed in this paper, such improvements can contribute to more sustainable and efficient urban logistics.

V. CONCLUSION

This paper introduces a cognitive decision-making framework for autonomous driving systems that aligns with subjective safety and promotes user trust, and proposes a methodology for investigating human-robot interaction. It is expected that by ensuring adaptability within strict safety boundaries, the framework will support wider acceptance of AD technologies. Its modular design will allow seamless integration across various AD systems, including mobile robots and autonomous vehicles. Implementation, testing, and application of the framework described in Section III, incorporating the optimal ranges for subjective safety factors from Sections I-A and II, are the essential steps. It is foreseen that the AD system's behaviour will be compared to assess whether it is perceived as safer with or without the proposed framework. Finally, subjective safety behaviour recommendations, regarding speed and distance, will be compiled and published for developers and researchers to integrate into their decision-making systems.

The following extensions are planned. In the proposed state, the framework makes decisions based on offline data. A natural extension is to incorporate sensor data online, to account for inter-personal differences and to adapt to the specific behaviour of interacting VRUs. A stress recognition system can be developed and trained based on empirical data from critical encounters during testing to extend the decision-making system with adaptive responses to external stress markers (based on body posture, eye tracking information, gesture recognition, etc.) as a part of the *Situation Interpretation* cluster. While the proposed framework will be developed and tested in simplified, controlled scenarios, real urban environments involve greater complexity, including multimodal traffic (diversity of traffic participants), variable infrastructure and environment conditions (cycle path width, weather conditions, etc.), and unpredictable events. This may limit the direct transferability of the identified safety parameters. Future work will therefore focus on a broader spectrum of environmental conditions and their effect on the interaction, along with the adaptation of the framework in progressively more complex and ecologically valid contexts.

REFERENCES

- [1] U. Plank-Wiedenbeck, O. Singler, R. Kramm, and F. Schrödel, "Co-operative and connected driving in urban environments," Bauhaus-Universität Weimar, Weimar, Germany, Tech. Rep., 2024.

- [2] M. Fuder, Y. Hagenbach, A. Huemer, A. Kläver, J. Krause, J. Kühl, P. Pez, and L. Schütte, "Mehr vielfalt im radverkehr - entdecken - gestalten - vermitteln," in *Forum Mobilität - Schriftenreihe des Instituts für Verkehrsmanagement der Ostfalia Hochschule für angewandte Wissenschaften Braunschweig/Wolfenbüttel*, ser. Forum Mobilität, J. Kühl, Ed., no. 1, 2023, p. 70.
- [3] J. Vanparijs, L. Int Panis, R. Meeusen, and B. de Geus, "Exposure measurement in bicycle safety analysis: A review of the literature," *Accident Analysis & Prevention*, vol. 84, pp. 9–19, 2015.
- [4] H. M. A. Aziz, N. N. Nagle, A. M. Morton, M. R. Hilliard, D. A. White, and R. N. Stewart, "Exploring the impact of walk-bike infrastructure, safety perception, and built-environment on active transportation mode choice: a random parameter model using new york city commuter data," *Transportation*, vol. 45, no. 5, pp. 1207–1229, 2018.
- [5] S. Daraei, K. Pelechrinis, and D. Quercia, "A data-driven approach for assessing biking safety in cities," *EPJ Data Science*, vol. 10, no. 1, p. 11, 2021.
- [6] S. Fuest, M. Batista, F. L. Berghoefer, M. Flessner, B. S. Grandhi, F. Spühler, M. Sester, and M. Vollrath, "I bet you feel safe! assessing cyclists' subjective safety by objective scores," *Journal of Urban Mobility*, vol. 4, p. 100066, 2023.
- [7] M. Sørensen and M. Mosslemi, "Subjective and objective safety: The effect of road safety measures on subjective safety among vulnerable road users." Institute of Transport Economics, Oslo, Norway, Tech. Rep. 1009/2009, 2009. [Online]. Available: <https://www.toi.no/getfile.php?mmfileid=11739>
- [8] K. Holländer, M. Colley, E. Rukzio, and A. Butz, "A taxonomy of vulnerable road users for hci based on a systematic literature review," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, ser. CHI '21. New York, NY, USA: ACM, 2021, p. 158.
- [9] T. Götschi, A. Castro, M. Deforth, L. Miranda-Moreno, and S. Zangenehpour, "Towards a comprehensive safety evaluation of cycling infrastructure including objective and subjective measures," *Journal of Transport & Health*, vol. 8, pp. 44–54, 2018.
- [10] S. Zug, N. Seyffer, M. Plank, B. Pfleging, F. Schrödel, and F. W. Siebert, "Identification of potential conflict zones between pedestrians and mobile robots in urban situations," in *Proceedings of the 2023 IEEE 28th International Conference on Emerging Technologies and Factory Automation (ETFA)*. New York, NY, USA: IEEE, 2023.
- [11] Q. Liu, X. Li, S. Yuan, and Z. Li, "Decision-making technology for autonomous vehicles: Learning-based methods, applications and future outlook," in *2021 IEEE International Intelligent Transportation Systems Conference (ITSC)*, 2021, pp. 30–37.
- [12] R. L. Sanders, "Perceived traffic risk for cyclists: The impact of near miss and collision experiences," *Accident Analysis & Prevention*, vol. 75, pp. 26–34, 2015.
- [13] S. Gössling and S. McRae, "Subjectively safe cycling infrastructure: New insights for urban designs," *Journal of Transport Geography*, vol. 101, p. 103340, 2022.
- [14] S. Rezwana, M. R. R. Shaon, N. Lownes, and E. Jackson, "Bridging the gap: understanding the factors affecting pedestrian safety perceptions in the age of driverless vehicles," *Traffic Safety Research*, vol. 9, p. e000080, 2025.
- [15] S. Thellman, F. Babel, and T. Ziemke, "Cycling with robots: How long-term interaction experience with automated shuttle buses shapes cyclist attitudes," in *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, ser. HRI '24. New York, NY, USA: ACM, 2024, pp. 1043–1047.
- [16] F. W. Siebert, J. Pickl, J. Klein, M. Rötting, and E. Roesler, "Let's not get too personal – distance regulation for follow me robots," in *HCI International 2020 – Late Breaking Posters*, C. Stephanidis, M. Antona, and S. Ntoa, Eds. Cham, Switzerland: Springer International Publishing, 2020, pp. 459–467.
- [17] B. Leichtmann and V. Nitsch, "How much distance do humans keep toward robots? literature review, meta-analysis, and theoretical considerations on personal space in human-robot interaction," *Journal of Environmental Psychology*, vol. 68, p. 101386, 2020.
- [18] Y. Strigina, F. Siebert, S. Tendulkar, V. Uppalapati, and F. Schrödel, "Human-robot interaction in urban areas – development and validation of acceptance proto-model in the scope of proximity theory," in *21. Konferenz für Angewandte Automatisierungstechnik in Lehre und Entwicklung an Hochschulen (AALE)*. Leipzig, Germany: Hochschule für Technik, Wirtschaft und Kultur Leipzig, 2025, pp. 377–385.
- [19] A. Gorrini, L. Crociani, G. Vizzari, and S. Bandini, "Stress estimation in pedestrian crowds: Experimental data and simulations results," *Web Intelligence*, vol. 17, no. 1, pp. 85–99, 2019.
- [20] G. V. Portnova, K. M. Liaukovich, L. N. Vasilieva, and E. I. Alshanskaiia, "Autonomic and behavioral indicators on increased cognitive loading in healthy volunteers," *Neuroscience and Behavioral Physiology*, vol. 53, no. 1, pp. 92–102, 2023.
- [21] A. Magliacano, S. Fiorenza, A. Estraneo, and L. Trojano, "Eye blink rate increases as a function of cognitive load during an auditory oddball paradigm," *Neuroscience Letters*, vol. 736, p. 135293, 2020.
- [22] B. Mahanama, Y. Jayawardana, S. Rengarajan, G. Jayawardana, L. Chukoskie, J. Snider, and S. Jayarathna, "Eye movement and pupil measures: A review," *Frontiers in Computer Science*, vol. 3, p. 733531, 2022.
- [23] Q. Zhao and C. Koch, "Learning a saliency map using fixated locations in natural scenes," *Journal of Vision*, vol. 11, no. 3, p. 9, 2011.
- [24] S. Ishaque, N. Khan, and S. Krishnan, "Trends in heart-rate variability signal analysis," *Frontiers in Digital Health*, vol. 3, p. 639444, 2021.
- [25] G. I. Christopoulos, M. A. Uy, and W. J. Yap, "The body and the brain: Measuring skin conductance responses to understand the emotional experience," *Organizational Research Methods*, vol. 22, no. 1, pp. 394–420, 2019.
- [26] S. Chen, J. Epps, and F. Chen, "A comparison of four methods for cognitive load measurement," in *Proceedings of the 23rd Australian Computer-Human Interaction Conference*, ser. OzCHI '11. New York, NY, USA: ACM, 2011, pp. 76–79.
- [27] E. Vaportzis, N. Georgiou-Karistianis, and J. C. Stout, "Dual task performance in normal aging: A comparison of choice reaction time tasks," *PLoS ONE*, vol. 8, no. 3, p. e60265, 2013.
- [28] J. Grimmer, L. Simon, and J. Ehlers, "The cognitive eye: Indexing oculomotor functions for mental workload assessment in cognition-aware systems," in *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, ser. CHI EA '21. New York, NY, USA: ACM, 2021, p. 428.
- [29] N. Akalin, A. Kristofferson, and A. Loutfi, "Do you feel safe with your robot? factors influencing perceived safety in human-robot interaction based on subjective and objective measures," *International Journal of Human-Computer Studies*, vol. 158, p. 102744, 2022.
- [30] M. Hoggemüller, M. Tomitsch, L. Hespahnol, T. T. M. Tran, S. Worrall, and E. Nebot, "Context-based interface prototyping: Understanding the effect of prototype representation on user feedback," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, ser. CHI '21. New York, NY, USA: ACM, 2021, p. 370.
- [31] O. Singler, L. Thiebes, M. Fedior, L. Disse, and U. Plank-Wiedenbeck, "Intersecting urban and traffic models for virtual reality environments with realistic traffic behaviour," in *12th International Conference "Environmental Engineering"*. Vilnius, Lithuania: Vilnius Gediminas Technical University, 2023.
- [32] D. Ullmann, J. Kreimeier, T. Götzelmann, and H. Kipke, "Bikevr: A virtual reality bicycle simulator towards sustainable urban space and traffic planning," in *Proceedings of Mensch Und Computer 2020*, ser. MuC '20. New York, NY, USA: ACM, 2020, pp. 511–514.
- [33] P. Mayring, *Qualitative Inhaltsanalyse. Grundlagen und Techniken*, 12th ed. Weinheim, Germany: Beltz, 2015.
- [34] C. Leys, C. Ley, O. Klein, P. Bernard, and L. Licata, "Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median," *Journal of Experimental Social Psychology*, vol. 49, no. 4, pp. 764–766, 2013.
- [35] R. Voßwinkel, M. Gerwien, A. Jungmann, and F. Schrödel, "Intelligent decision-making and motion planning for automated vehicles," in *Autonomous Driving and Advanced Driver-Assistance Systems (ADAS): Applications, Development, Legal Issues, and Testing*, 1st ed., L. Joseph and A. K. Mondal, Eds. Boca Raton, FL, USA: CRC Press, 2021, pp. 3–35.
- [36] Q. Yang, T. D. Simão, S. H. Tindemans, and M. T. J. Spaan, "Wcsac: Worst-case soft actor critic for safety-constrained reinforcement learning," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 12, pp. 10639–10646, 2021.
- [37] T. Haarnoja, A. Zhou, K. Hartikainen, G. Tucker, S. Ha, J. Tan, V. Kumar, H. Zhu, A. Gupta, P. Abbeel, and S. Levine, "Soft actor-critic algorithms and applications," 2019. [Online]. Available: <https://arxiv.org/abs/1812.05905>
- [38] K. Othman, "Public acceptance and perception of autonomous vehicles: a comprehensive review," *AI Ethics*, vol. 1, no. 3, pp. 355–387, 2021.